Regime-Conditioned Trade Flow Imbalance and Adverse Selection in ES Futures

Gabriel Ungab

Georgia Institute of Technology

May 2026

Abstract. This paper examines whether conditioning Trade Flow Imbalance (TFI) on a real-time informed trading regime detector amplifies its predictive power for forward returns in E-mini S&P 500 futures. The regime detector is a two-component multiplicative composite of Kyle's lambda and trade arrival rate, constructed from 58.8 million trades across 169 trading days on CME Globex and validated on a held-out 46-day out-of-sample period. The primary finding is a null result: regime-conditioned TFI does not significantly predict forward returns at any horizon from $T+1$ through $T+15$ ($\hat{\beta}_3 = 0.000371$, $p = 0.234$). The detector validation separately establishes 2.278× within-bar price impact amplification in high-regime bars ($p < 0.001$). If this amplification is genuine rather than circularity-driven, the null forward-return result reflects within-bar information incorporation rather than an absence of signal; a follow-on study with an orthogonal LOB-based detector will determine which interpretation is warranted. A key secondary finding qualifies this conclusion: restricting the regression to bars where the lambda estimation window is most stable produces a statistically significant regime-TFI interaction ($\hat{\beta}_3 = 0.001016$, $p = 0.033$) that is 2.7× the full-sample estimate, suggesting the detector's signal quality depends critically on estimation reliability. Three caveats bound this finding: the stability threshold is derived post-hoc from the in-sample distribution; the effect does not replicate in the held-out out-of-sample period; and the stability metric is itself partially circular with the regime detector. The finding is treated as a genuine in-sample result motivating a specific future research direction rather than a deployable signal. A central methodological contribution is the explicit documentation of two circularity layers inherent in trades-only regime detection and the analysis of their directional effects on all estimates. Both layers act to bias $\hat{\beta}_3$ upward toward the alternative hypothesis, a direction confirmed by a permutation simulation (1,000 datasets under H_0, seed = 42) that places the observed $\hat{\beta}_3 = 0.000371$ at the 85.4th percentile of the null distribution (simulation p-value = 0.146), making the primary null result a conservative bound on the true causal effect.

JEL Classification: G12, G14, C58

Keywords: market microstructure, adverse selection, informed trading, order flow imbalance, Kyle's lambda, ES futures, regime detection

1. Introduction

Market makers in high-frequency equity index futures face a fundamental adverse selection problem. When a counterparty's order flow reflects private information about future price direction, market makers who fail to detect it provide liquidity at stale quotes and absorb losses from informed traders. Managing this exposure requires identifying, in real time, whether the order flow arriving in each bar is more likely to be informed or uninformed. The theoretical frameworks of Glosten and Milgrom (1985) and Kyle (1985) formalize this problem: market makers update quotes in proportion to their inference about order flow informativeness, and the price impact of a trade — Kyle's lambda — is the empirical signature of this updating process. The practical question is whether lambda, observed from trades-only data, can be combined with other market characteristics to construct a real-time regime indicator that reliably identifies elevated adverse selection.

This paper asks whether conditioning Trade Flow Imbalance (TFI) on such a regime indicator amplifies its predictive power for next-bar forward returns in E-mini S&P 500 (ES) futures. TFI measures the normalized directional imbalance between buyer- and seller-initiated volume within each 1-minute bar, a signal that reflects the net pressure of order flow but mixes informed and uninformed activity unconditionally. The regime detector is a two-component multiplicative composite of Kyle's lambda and trade arrival rate (TAR), constructed from 58.8 million RTH trades across 169 trading days on CME Globex. The primary test is a regression of next-bar log returns on TFI, RegimeScore, and their interaction, estimated with Newey-West HAC standard errors on the full in-sample period (2025-05-01 through 2025-12-30) and validated on a held-out out-of-sample period (2026-01-02 through 2026-03-06).

The primary finding is a null result. Regime-conditioned TFI does not significantly predict forward returns at any horizon from $T{+}1$ through $T{+}15$ ($\hat{\beta}_3 = 0.000371$, $p = 0.234$ at $T{+}1$), implying that ES futures incorporate regime-conditioned order flow information within one 1-minute bar and leave no exploitable residual predictability at any tested horizon. A key secondary finding qualifies this conclusion: when the regime detector's lambda estimation window is stable — defined as the bottom tercile of rolling signed-flow standard deviation — the regime-TFI interaction is statistically significant ($\hat{\beta}_3 = 0.001016$, $p = 0.033$) and 2.7× the full-sample estimate. This suggests the detector's signal quality depends critically on estimation reliability, and motivates a confirmatory test with an orthogonal limit order book detector as the primary direction for future research.

A central methodological challenge in this analysis is that with trades-only data, a fully orthogonal regime detector is impossible: Kyle's lambda and TFI share aggressor-side signed order flow as a common input. This paper explicitly documents two layers of circularity — at the detector construction level and at the regression interaction level — and analyzes their directional effects on all estimates throughout. The circularity biases $\hat{\beta}_3$ upward toward the alternative hypothesis, making a null result more informative: it holds even under estimation conditions that favor finding a positive coefficient. This accounting of what the data can and cannot identify is a primary methodological contribution of the paper.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the data and sample construction. Section 4 presents the signal construction, regime detector design, exclusion windows, detector validation, and empirical specification. Section 5 reports the primary results and additional tests. Section 6 presents robustness checks. Section 7 discusses market maker implications. Section 8 outlines two directions for future research.

2. Literature Review

The theoretical foundation of this paper rests on the adverse selection frameworks of Kyle (1985) and Glosten and Milgrom (1985). Kyle (1985) models a market in which a single informed trader submits orders strategically to minimize price impact, and derives lambda — the price impact per unit of signed order flow — as the equilibrium measure of information asymmetry. Glosten and Milgrom (1985) model competitive market makers who update bid and ask quotes in response to the probability that any incoming order is informed, establishing that the optimal spread is proportional to adverse selection risk. Together, these frameworks motivate the regime detector design: a period of elevated lambda, confirmed by active market participation, is the empirical signature of the informed trading conditions both frameworks identify as generating adverse selection. The detector is not a loose empirical proxy; it is a direct operationalization of the quantities these models define.

The empirical antecedent of TFI is the order flow imbalance (OFI) measure of Cont et al. (2014), who show that OFI constructed from limit order book events predicts short-term price changes in equity markets. This paper uses a trades-only analog (signed volume imbalance at the aggressor side) and replicates the finding that unconditional order flow imbalance predicts forward returns ($\hat{\beta}_1 = 0.000637$, $p < 0.001$). The central question of this paper is whether regime conditioning amplifies this unconditional predictability. It does not — but the stable regime conditions analysis identifies the specific circumstances under which a statistically detectable amplification does emerge, providing a bridge from Cont et al.'s unconditional finding to a conditional one that may be accessible with richer data.

The choice of detector components is grounded in Ahern (2018) and Dufour and Engle

(2000). Ahern (2018) evaluates which trades-only proxies for informed trading most accurately identify actual insider trading episodes using a unique sample of illegal trades, finding that the most accurate proxies — TFI variants and order flow autocorrelation — are informationally circular with TFI itself. This directly motivates the use of lambda and TAR as the least circular available detector: they capture price impact and market activity without using TFI's own directional content as an input. Roll (1984) was considered as a third component — its serial covariance estimator measures bid-ask tightness and could confirm that elevated lambda reflects genuine price impact rather than wide spreads — but was excluded because it is non-orthogonal to lambda and fails precisely in the one-sided markets that represent the highest-lambda informed trading episodes of greatest theoretical interest.

The most prominent trades-only informed-trading proxy not used in this paper is VPIN (Volume-Synchronized Probability of Informed Trading), introduced by Easley et al. (2012). VPIN was considered as both an alternative regime detector and a robustness check and was excluded for two reasons. First, VPIN is constructed from signed volume imbalance — the same directional content as TFI — making it more circular with TFI than lambda, which at minimum incorporates price impact as an independent input. Second, because the primary result is null, a VPIN robustness check would confirm a null result under a more circular detector, adding no inferential value. The exclusion is deliberate: the detector design goal was to minimize circularity with TFI, not to benchmark against existing proxies that inherit the same limitation.

Dufour and Engle (2000) provide direct theoretical grounding for TAR: they show that inter-trade duration is informative about information arrival, with faster trading associated with informed episodes. This motivates including TAR as a real-time proxy for market participation intensity and supports its role as a depth confirmation for elevated lambda rather than as an independent adverse selection signal. The use of TAR as a hierarchical check on lambda connects to the broader microstructure literature surveyed in O'Hara (1995): informed trading episodes are characterized jointly by elevated price impact and active participation, not by price impact alone.

This paper contributes to the microstructure literature in three ways. First, it introduces a multiplicative regime detector that enforces the theoretical hierarchy between lambda and TAR — lambda provides the primary adverse selection signal and TAR confirms or suppresses it — rather than treating the two components as symmetric additive inputs. Second, it provides a systematic documentation of two distinct circularity layers in trades-only regime detection and analyzes their directional effects on all downstream estimates, addressing a methodological gap in the empirical literature on informed trading proxies. Third, the stable regime conditions finding — that the regime-TFI interaction is statistically significant when lambda estimation is most reliable — generates a specific, falsifiable prediction for future research with limit order book data: an orthogonal LOB-based detector operating under stable liquidity conditions should produce significant regime-conditioned forward return predictability, or confirm that the in-sample finding is circularity-driven.

3. Data

3.1. Source and Instrument

This study uses trade-level data from CME Globex obtained via Databento (GLBX.MDP3, Trades schema). The data provides a complete record of every executed trade in ES futures at nanosecond timestamp precision, including execution price, contract size, and aggressor side (B = buyer, A = seller, N = unknown). Analysis is restricted to Regular Trading Hours: 9:30 AM to 4:00 PM Eastern time. The overnight Globex session is excluded due to its structurally different liquidity profile. Calendar spread instruments (e.g. ESM5-ESU5), which appeared in the raw feed with spread prices rather than outright futures prices, were removed by filtering any symbol containing a hyphen.

3.2. Sample Period and Cleaning

The in-sample period covers 2025-05-01 through 2025-12-30. A held-out out-of-sample period spanning 2026-01-02 through 2026-03-06 is reserved exclusively for final validation and is not used in any model estimation or specification search.

The in-sample period (May–December 2025, 169 trading days) was determined by data availability and cost. Trade-level data was purchased for this study period only; pre-May 2025 GLBX.MDP3 data was not acquired due to cost constraints and therefore was not available when the formal analysis was conducted. The OOS window (January–March 2026, 46 trading days) is the next available period after the in-sample window and was held out without any access during model development. A longer history or multiple rolling OOS windows would be preferable and is a direction for future work; the conclusions of this paper are therefore conditional on the 2025–2026 sample and the market conditions prevailing during it.

Four trading days were excluded due to anomalously low activity relative to the sample mean of 348,369 trades per day: Memorial Day (2025-05-26, 14,711 trades), Independence Day (2025-07-04, 13,367 trades), Labor Day (2025-09-01, 11,431 trades), and the half-session preceding Thanksgiving (2025-11-27, 9,258 trades). A natural gap separates these four days from all other sessions, including recognized reduced-activity days such as Juneteenth (51,784 trades) which are retained as legitimate trading sessions.

Price outliers were removed using a rolling five-standard-deviation filter with a window of

100 trades, eliminating 114,413 records (≈0.19% of the raw sample). An additional 2,856,579 records share nanosecond timestamps with at least one other trade; inspection confirmed these as legitimate multi-level fills and all are retained. The final clean in-sample dataset contains ≈58.8 million RTH trades across 169 trading days.

3.3. Descriptive Statistics

Table 1 presents daily summary statistics across the 169-day in-sample period.

Table 1. Daily Summary Statistics, In-Sample Period (May–December 2025, N = 169 trading days)

Statistic	Mean	Std	Min	Max
Trade count	348,369	142,128	51,784	1,015,024
Total volume (contracts)	1,070,364	335,016	121,454	2,509,428
Avg. trade size	3.17	0.33	2.35	3.92
Price range (pts)	121.4	38.8	30.5	298.5
Buy/sell volume ratio	1.003	0.022	0.942	1.096
Realized vol (per min)	0.000894	0.001429	0.000125	0.006346

Buyer- and seller-initiated trades account for 49.91% and 50.09% of volume respectively, with unknown aggressor side comprising a negligible 0.000003% of records. Individual trade sizes are highly right-skewed; the median is 1 contract and the 75th percentile is 3 contracts, while the maximum reaches 1,968 contracts. Daily volume was relatively stable at ≈1 million contracts per day from May through September 2025, becoming more elevated and volatile from October through December, suggesting two distinct activity regimes within the sample period (Figure 1).

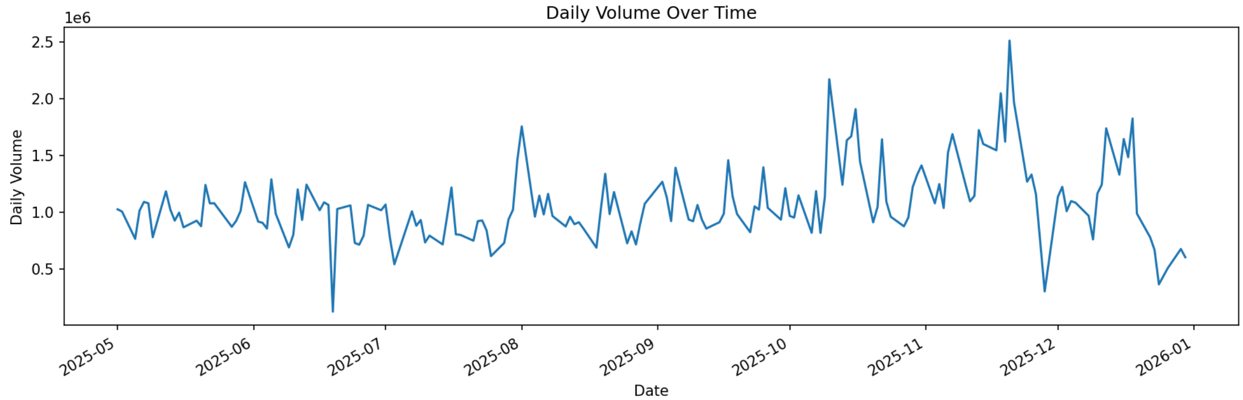

Figure 1. Daily total volume across the 169-day in-sample period. Two distinct activity regimes are visible: a stable period from May through September 2025 and a more elevated, volatile period from October through December 2025.

3.4. Intraday Patterns

Intraday volume exhibits a modified inverse-J shape across the trading day, peaking at the open, declining steadily through the morning, remaining low through midday, and surging sharply at the close, reaching ≈52,000 contracts per minute in the final minute of trading, driven by market-on-close (MOC) order flow (Figure 2). Trade arrival rate follows the same pattern, with ≈4,000 trades per minute at the open and ≈7,500 at the close, indicating that the close-of-day spike reflects both more trades and larger average trade sizes.

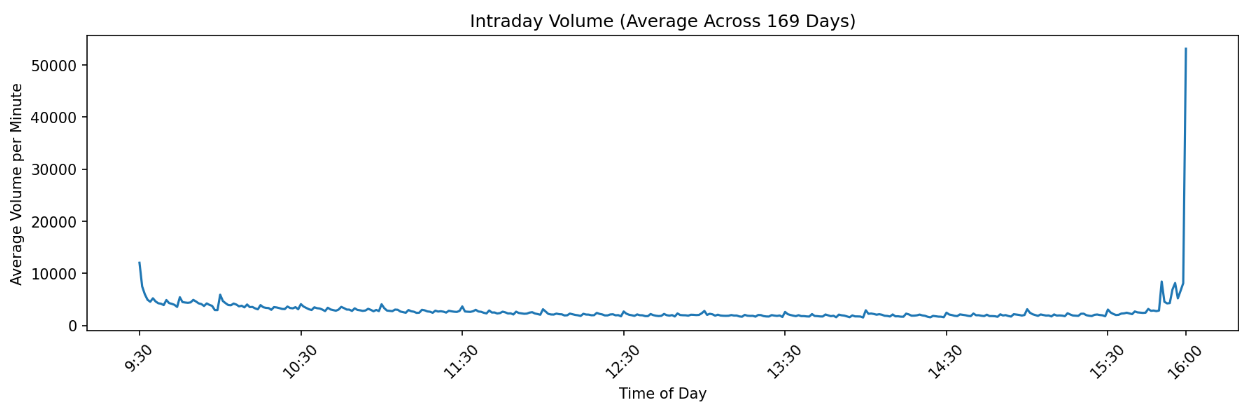

Figure 2. Average intraday volume by minute, averaged across 169 trading days. The modified inverse-J pattern peaks at the open and surges at the close, driven by MOC order flow.

Intraday realized volatility exhibits a distinctly different pattern (Figure 3). Volatility spikes sharply at the open (≈0.0053 per minute, ≈3× the midday average) as overnight information is incorporated into prices, then declines rapidly and remains flat through midday and into the close, with no corresponding spike despite the sharp increase in volume. This divergence is consistent with MOC-driven close activity — uninformed, publicly announced order flow that generates no adverse selection and therefore minimal price impact — in contrast to the open, where informed order flow drives both high volume and high volatility through adverse selection.

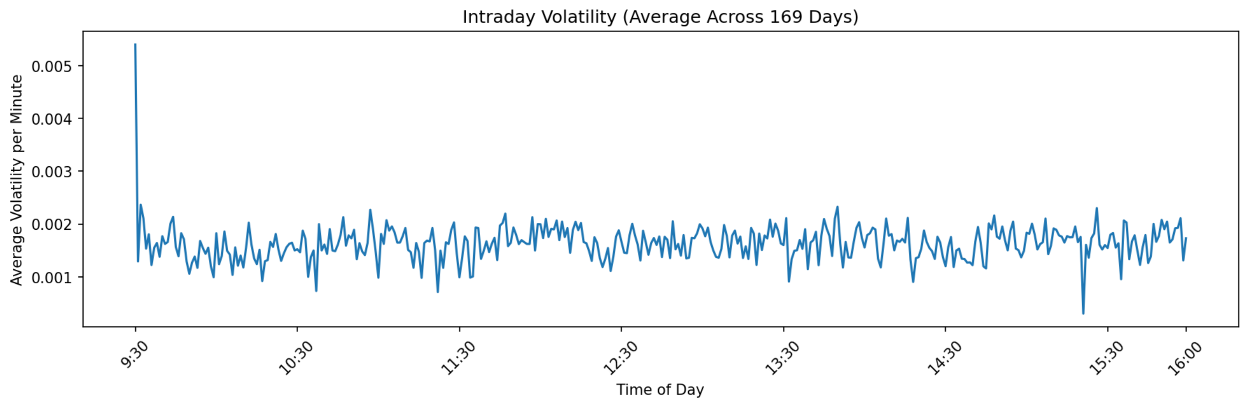

Figure 3. Average intraday realized volatility by minute, averaged across 169 trading days. Volatility is highest at the open and flat at the close despite high MOC volume, consistent with uninformed close-of-day execution.

4. Signal Construction and Empirical Design

4.1. Trade Flow Imbalance

The primary signal is Trade Flow Imbalance (TFI), defined as the normalized difference between buyer- and seller-initiated volume within each 1-minute bar:

$$\mathrm{TFI}_t = \frac{\mathrm{BuyVol}_t - \mathrm{SellVol}_t}{\mathrm{BuyVol}_t + \mathrm{SellVol}_t} \tag{1}$$

where BuyVol_t and $\mathrm{SellVol}_t$ are the total contract size of buyer-initiated (aggressor side B) and seller-initiated (aggressor side A) trades respectively within bar t. Thus $\mathrm{TFI}_t \in [-1, 1]$, with positive values indicating net buying pressure and negative values indicating net selling pressure.

Unconditionally, TFI mixes informed and uninformed order flow. Index fund rebalancing, inventory management by market makers, and retail liquidity trades all contribute to TFI without carrying information about future price direction. The analysis therefore conditions TFI on a regime indicator designed to isolate periods where order flow is more likely to reflect private information.

4.2. Informed Trading Regime Score

The informed trading regime score, $\text{RegimeScore}_t \in [0, 1]$, is a continuous measure of the probability that bar t occurs during a period of elevated information asymmetry. Kyle's lambda is chosen as the primary component because it is the most direct available measure of adverse selection intensity: it captures price impact per unit of signed order flow in real time from trades-only data, which is precisely the quantity the Glosten and Milgrom (1985) and Kyle (1985) frameworks identify as the signature of informed trading. However, high lambda alone is insufficient; it can be elevated for five distinct reasons besides genuine informed trading: thin markets or low depth, market stress or sudden uncertainty, market maker inventory pressure, mechanical order clustering, and contract roll proximity. The regime detector is designed to address as many of these as possible given the constraints of trades-only data.

Kyle's lambda measures price impact per unit of signed order flow, estimated from a rolling OLS regression of bar-level price changes on signed order flow over a 30-bar window updated each minute. Specifically, ΔPrice_t is the close-to-close price change within bar t (in index points), and SignedFlow_t is the net buyer-initiated minus seller-initiated contracts within bar t, yielding one observation per bar in the OLS. The 30-bar window is pre-specified from theory: it is consistent with rolling lambda windows used in the microstructure literature, provides sufficient observations for stable OLS estimation, and remains short enough to be reactive to intraday regime shifts. The window was fixed before any regressions were run and no sensitivity analysis was conducted; the robustness of results to alternative window lengths (e.g., 15 or 60 bars) is a direction for future work.

Trade arrival rate (TAR) measures trades per minute over a 5-minute rolling window and functions as a depth confirmation for elevated lambda. High arrival rate confirms that many market participants are actively transacting, ruling out thin markets and low depth as the source of elevated lambda. When TAR is low, RegimeScore is suppressed even if lambda is elevated, preventing misclassification of illiquid episodes as informed.

Addressing the five sources of elevated lambda. One of the five non-informed sources is addressed by TAR directly: thin markets and low depth produce low TAR, suppressing RegimeScore even when lambda is elevated. Two more are addressed by the exclusion windows described in Section 4.3: market stress from scheduled macro announcements and contract roll proximity. Two remain unaddressed as an irreducible limitation of trades-only data; market maker inventory pressure and mechanical intraday order clustering outside exclusion windows can produce elevated lambda that the detector cannot distinguish from genuine informed flow.

Multiplicative combination. The two components are combined using a multiplicative formulation that implements the hierarchical structure described above. Each component is first standardized via a rolling z-score:

$$z_{\lambda,t} = \frac{\lambda_t - \mu_\lambda}{\sigma_\lambda} \tag{2}$$

$$z_{\mathrm{TAR},t} = \frac{\mathrm{TAR}_t - \mu_{\mathrm{TAR}}}{\sigma_{\mathrm{TAR}}} \tag{3}$$

Each z-score is then mapped to $(0, 1)$ via the logistic function, and the two logistic outputs are multiplied:

$$\mathrm{RegimeScore}_t \;=\; \mathrm{logistic}(z_{\lambda,t}) \times \mathrm{logistic}(z_{\mathrm{TAR},t})\,, \qquad \mathrm{logistic}(x) = \frac{1}{1 + e^{-x}} \tag{4}$$

This formulation ensures RegimeScore is high only when both components are simultaneously elevated. Rolling standardization ensures the score is always relative to recent market conditions rather than the full sample distribution, allowing it to adapt to the two distinct activity regimes identified in Section 3. All rolling estimates use only past data; no lookahead bias is introduced.

Design choice—Roll spread exclusion. The Roll (1984) spread estimator was considered and excluded as a third component for two reasons. First, Roll is non-orthogonal to lambda; both spike simultaneously at the same directional price episodes, making Roll redundant rather than independently informative. Second, Roll fails precisely in one-sided markets — the high-lambda informed trading conditions of greatest theoretical interest — because the bid-ask bounce required for the serial covariance estimator disappears when prices move consistently in one direction.

A fundamental constraint of this design is that lambda and TFI share a common input; both are derived from aggressor-side signed order flow. Any regime detector constructed from trades-only data inherits this circularity at the construction level. The implications for all downstream tests are discussed in Section 4.5.

4.3. Exclusion Windows

Two sources of elevated lambda that do not reflect genuine informed trading — post-announcement mechanical flow and MOC order flow, and contract roll illiquidity — are addressed through targeted exclusion windows. $\mathrm{RegimeScore}_t$ is set to zero in all excluded bars regardless of indicator values.

Scheduled macro announcements (FOMC decisions, CPI releases, NFP releases) trigger a 30-minute post-announcement exclusion window. These events drive directional order flow from participants responding to the same public information signal rather than private information, and their post-announcement windows represent the period of greatest mechanical lambda inflation. Pre-announcement windows are retained; informed traders with early signal access may be active before announcements, and excluding these bars would discard potentially genuine informed trading episodes.

Descriptively, 2 of the 6 in-sample FOMC pre-announcement windows (June 18 and September 17, 2025) fall within contract roll exclusion windows, setting RegimeScore to zero by construction in those 60 bars. For the remaining 4 events (May 7, July 30, October 29, December 10), the 30-minute pre-announcement windows contain 120 bars with mean RegimeScore = 0.275, compared to the full-sample mean of 0.215, directionally consistent with elevated adverse selection before announcements, though 4 events are insufficient for inference. Mean |TFI| across all 6 pre-announcement windows (180 bars) is 0.097 versus 0.090 full-sample, a modest elevation unaffected by roll contamination. Both comparisons are descriptive only.

The final 10 minutes of each session (15:50–16:00 ET) are excluded due to domination by MOC order flow — mechanical, publicly announced execution that generates high volume without adverse selection, as documented in Section 3.4. Contract roll dates and the three preceding trading days are excluded to remove the structural illiquidity and mechanical volume migration accompanying front-month contract expiration.

4.4. Regime Detector Validation

Prior to testing TFI predictability, the regime detector is validated through a formal statistical test of its core claim: high-regime bars have a significantly stronger within-bar TFI-return relationship than low-regime bars. This is the mechanism the detector is designed to identify: elevated adverse selection produces stronger price impact per unit of order flow within the same bar, per Kyle (1985). A significant amplification confirms the detector correctly identifies periods of elevated within-bar price impact.

The validation regresses contemporaneous returns on TFI, a high-regime dummy, and their interaction:

$$R_t = \alpha + \beta_1 \mathrm{TFI}_t + \beta_2 \mathbf{1}[\mathrm{RegimeScore}_t > 0.5] + \beta_3 \Big(\mathrm{TFI}_t \times \mathbf{1}[\mathrm{RegimeScore}_t > 0.5]\Big) + \beta_5 \mathrm{TFI}_{t-1} + \varepsilon_t \tag{5}$$

The interaction coefficient $\hat{\beta}_3 = 0.001525$ ($z = 9.496$, $p < 0.001$). The within-bar TFI-return slope is 0.001193 in low-regime bars and 0.002717 in high-regime bars, a 2.278× amplification that is strongly statistically significant. This confirms that the detector reliably identifies periods in which order flow has disproportionately larger within-bar price impact, which is the direct empirical implication of the Kyle (1985) adverse selection framework.

The amplification figure is a conditional association under confounded conditions, not a causal estimate; TFI and R_t share the same bar's trades. However, confounding affects both the high-regime and low-regime slope equally through β_1. The interaction term β_3 specifically measures the *additional* TFI-return slope in high-regime bars relative to low-regime bars, and this differential amplification cannot be explained by confounding alone.

A pre-announcement validation was considered — testing whether RegimeScore is elevated in the 30-minute window before macro announcements, when informed traders with early signal access should be active. This test is infeasible with RTH-only data; CPI and NFP releases occur at 8:30 AM before market open, leaving only 6 observable FOMC pre-announcement windows in the in-sample period, an insufficient sample for a meaningful test.

4.5. Empirical Design

Researcher degrees of freedom disclosure. In the interest of transparency, the following specification choices were fixed before any regressions were run: 1-minute bar aggregation (standard in the microstructure literature); RegimeScore threshold of 0.5 (natural midpoint of the $[0, 1]$ logistic output); 30-bar lambda estimation window (pre-specified from theory); 5-minute TAR window (chosen to be shorter than the lambda window); HAC maxlags $= 5$ (standard for 1-minute return regressions); and the 30-minute post-announcement exclusion window (standard in the event study literature). One choice was made before formal regressions were run but after initial implementation: the multiplicative (rather than additive) combination of lambda and TAR was adopted after conceptual analysis of the hierarchical structure, with no predictability results observed under the additive formulation before the switch. One choice was determined after seeing results and is explicitly flagged as post-hoc: the bottom-tercile cut for stable-regime conditions in Section 5.5, acknowledged there as such.

All regressions use the following primary specification unless otherwise noted:

$$R_{t+1} = \alpha + \beta_1 \mathrm{TFI}_t + \beta_2 \mathrm{RegimeScore}_t + \beta_3(\mathrm{TFI}_t \times \mathrm{RegimeScore}_t) + \beta_4 R_t + \beta_5 \mathrm{TFI}_{t-1} + \varepsilon_t \quad (6)$$

β_1 captures the unconditional TFI effect when RegimeScore $= 0$. β_2 captures any return-level difference across regime states independent of TFI. β_3 is the primary coefficient of interest; the null hypothesis is H_0: $\beta_3 = 0$ against H_1: $\beta_3 > 0$. β_4 and β_5 control for return autocorrelation and TFI persistence. All regressions use Newey-West HAC standard errors with maxlags $= 5$.

The regression sample contains $N = 55{,}634$ bars after dropping rows arising from rolling warmup (the first 30 bars of each session), day boundary nulling (the first bar of each day whose return spans the overnight gap), and the forward return edge (the final bar of each day). High-regime bars (RegimeScore > 0.5) account for 12.1% of the regression sample.

Circularity and estimation bias. Lambda and TFI share aggressor-side signed order flow as a common input — the first layer of circularity, irreducible with trades-only data. A second layer operates at the regression interaction level: high-|TFI| bars mechanically have elevated RegimeScore because high signed flow simultaneously elevates lambda under the multiplicative formulation, inflating the interaction term TFI $\times$ RegimeScore. This second layer is confirmed by the monotonic quintile pattern documented in Section 5.4.

Both layers act to bias $\hat{\beta}_3$ upward toward H_1. The direction of this bias is confirmed by a permutation simulation. Forward returns are permuted 1,000 times (seed = 42) while preserving the joint distribution of TFI and RegimeScore; the full primary regression pipeline is applied to each permuted dataset and $\hat{\beta}_3$ is recorded. Under this constructed H_0, the null distribution has a positive mean (0.000011), confirming that mechanical co-elevation shifts the sampling distribution of $\hat{\beta}_3$ above zero even when forward returns carry no information. The observed $\hat{\beta}_3 = 0.000371$ falls at the 85.4th percentile of this null distribution (simulation p-value $= 0.146$), consistent with a null result under modest upward bias. On the stable-conditions subsample ($N = 18{,}355$), the simulation null mean is 0.000000 and the observed $\hat{\beta}_3 = 0.001016$ falls at the 98.5th percentile (simulation p-value $= 0.015$), indicating the stable-conditions in-sample result is not a mechanical artifact. Its out-of-sample non-replication therefore reflects a sample-specific phenomenon rather than a measurement bias. Taking the bias direction as confirmed, a null result under upward bias implies the true β_3 is no larger than the estimated value, making the primary efficiency finding a conservative bound. Figure A5 in Appendix A displays both null distributions with the observed values marked.

5. Results

5.1. Forward Return Predictability — T+1

The primary test asks whether regime-conditioned TFI predicts returns at the one-bar horizon. R_{t+1} is unknown when TFI_t and RegimeScore_t are observed. RegimeScore_t is contemporaneous with TFI_t but fully predetermined relative to R_{t+1}; the next bar's return is not in lambda's estimation window and is not known when the signal is recorded. This regression is non-confounded in the sense defined in Section 4.5.

Table 2. Primary Regression: R_{t+1} ($N = 55{,}634$; Newey-West HAC, maxlags $= 5$)

Variable	Coefficient	z-stat	p-value
Constant	-4.46×10^{-6}	-0.597	0.550
TFI_t (β_1)	0.000637	5.886	< 0.001***
RegimeScore_t (β_2)	1.75×10^{-5}	0.753	0.451
$\text{TFI}_t \times \text{RegimeScore}_t$ (β_3)	0.000371	1.191	0.234
R_t (β_4)	-0.488	-53.865	< 0.001***
TFI_{t-1} (β_5)	-4.96×10^{-5}	-0.902	0.367
$R^2 = 0.236$			

$\hat{\beta}_3 = 0.000371$, $p = 0.234$. H_0 is not rejected. The contemporaneous validation (Section 4.4) establishes that the regime detector identifies elevated within-bar price impact ($2.278\times$ amplification, $p < 0.001$). This within-bar amplification, if genuine and not purely circularity-driven, implies that regime-conditioned order flow information acts on prices within the bar rather than across bars, making the null forward-return result a consequence of within-bar information incorporation rather than an absence of signal. The extension's orthogonal LOB-based detector test will determine whether this interpretation is warranted. The unconditional TFI effect ($\hat{\beta}_1 = 0.000637$, $p < 0.001$) replicates the Cont et al. (2014) finding that order flow imbalance predicts short-term returns; this effect does not survive round-trip transaction costs of approximately 0.774 basis points and carries no directional trading application. The regime does not amplify this unconditional predictability; $\hat{\beta}_3$ is indistinguishable from zero. The dominant source of return predictability is the mean-reversion control R_t ($\hat{\beta}_4 = -0.488$, $p < 0.001$), capturing bid-ask bounce; $R^2 = 0.236$ is driven almost entirely by this term. Without the mean-reversion control, $R^2 = 0.000275$, confirming that the regime-TFI interaction accounts for essentially none of the return variance.

The quintile interaction analysis in Section 5.4 confirms this null result is not an artifact of the linearity assumption in the interaction specification. Figure A1 in Appendix A provides a visual analog of this regression, showing TFI versus one-bar-ahead log return stratified by regime state. Notably, the high-regime panel shows a slightly negative OLS slope (-4.61×10^{-4}) while the low-regime panel shows a slightly positive slope ($+1.07 \times 10^{-4}$); neither is economically distinguishable from zero at this scale, and the sign reversal in the high-regime panel is consistent with the mechanical co-elevation argument: high-|TFI| bars are disproportionately classified as high-regime through the circularity channel, inflating the regime label on bars where TFI ran against the subsequent return.

5.2. Contemporaneous Characterization — $T+0$

A separate analysis characterizes the regime-conditioned relationship between TFI and same-bar returns. Because R_t and RegimeScore_t share bar t's time window, RegimeScore is lagged by one bar (RegimeScore_{t-1}) to remove regime-level simultaneity:

$$R_t = \alpha + \beta_1 \text{TFI}_t + \beta_2 \text{RegimeScore}_{t-1} + \beta_3(\text{TFI}_t \times \text{RegimeScore}_{t-1}) + \beta_5 \text{TFI}_{t-1} + \varepsilon_t \quad (7)$$

A residual confounding channel remains irreducible; TFI_t and R_t are both constructed from bar t's trades, making their relationship partially mechanical with trades-only data. This regression is a specification sensitivity result — not a predictive test — and is presented as such.

Table 3. Contemporaneous Regression: R_t with Lagged RegimeScore_{t-1} ($N = 55{,}634$)

Variable	Coefficient	z-stat	p-value
Constant	-8.16×10^{-6}	-1.214	0.225
TFI_t (β_1)	0.001200	10.238	< 0.001***
RegimeScore_{t-1} (β_2)	3.53×10^{-5}	1.680	0.093
$\text{TFI}_t \times \text{RegimeScore}_{t-1}$ (β_3)	0.000617	1.761	0.078
TFI_{t-1} (β_5)	-0.000105	-1.769	0.077
$R^2 = 0.008$			

$\hat{\beta}_3 = 0.000617$, $p = 0.078$. The interaction is not significant at conventional thresholds. The lag-1 autocorrelation of RegimeScore is 0.806 under the multiplicative formulation, meaning lagging by one bar removes approximately 19% of the regime-level circularity. The $p = 0.078$ result is therefore best interpreted as predominantly driven by residual confounding through regime autocorrelation, with a non-trivial but unquantifiable secondary contribution from genuine predictive signal. No empirically grounded market maker calibration can be derived from this result.

5.3. Horizon Analysis

The primary specification is repeated at cumulative 5-bar and 15-bar forward horizons. Cumulative log returns are computed within each trading day only; cross-day windows are nulled to avoid overnight return contamination.

Table 4. Horizon Analysis: $\hat{\beta}_3$ at T+5 and T+15

Horizon	$\hat{\beta}_3$	z-stat	p-value	N
T+5	0.000090	0.256	0.798	54,789
T+15	0.000081	0.212	0.833	53,099

No regime interaction emerges at any horizon. The unconditional TFI effect ($\hat{\beta}_1 \approx$ 0.000696–0.000719, $p < 0.001$) persists at both horizons but does not vary with regime. Information incorporation in ES futures is complete within one 1-minute bar in high-regime conditions, leaving no residual regime-conditioned predictability at 5 or 15 bar horizons.

5.4. TFI Quintile Interaction

The continuous regime interaction term is replaced with four quintile dummy interactions, allowing the regime amplification to vary freely across the TFI distribution. Quintile 3 (moderate TFI, the theoretically predicted zone of informed order flow concentration under Kyle 1985) is the omitted reference category. A Bonferroni correction is applied across the five simultaneous tests ($\alpha = 0.01$ per test, family-wise $\alpha = 0.05$).

Table 5. TFI Quintile Interaction: $\hat{\beta}$ by Quintile (Reference: Q3; Bonferroni threshold $p < 0.01$)

Quintile	Coefficient	z-stat	p-value	Bonferroni
Q1 (most negative TFI)	−0.000020	−0.362	0.717	—
Q2	−0.000038	−0.786	0.432	—
Q4	0.000081	1.684	0.092	—
Q5 (most positive TFI)	0.000126	2.076	0.038	—

No quintile interaction survives Bonferroni correction. The null result in the primary regression is not a consequence of imposing linearity on the regime-TFI relationship. The monotonic pattern — coefficients increasing from Q1 through Q5 — is inconsistent with the Kyle (1985) prediction of hump-shaped concentration in moderate quintiles and is more consistent with mechanical co-elevation: higher absolute TFI implies higher signed order flow, which simultaneously elevates lambda and therefore RegimeScore, inflating the interaction term in high-TFI bars. This confirms the second layer of circularity described in Section 4.5.

5.5. Midday Subsample and Stable Regime Conditions

The regime detector's signal quality depends on the reliability of its component estimates. Kyle's lambda requires a full 30-bar estimation window of stable same-session data, and the

TAR z-score is most informative when baseline activity is low. Both conditions are best satisfied during the midday portion of the trading session (11:00–13:00 ET), after the open's elevated and noisy order flow has subsided and before close-of-day dynamics begin. This window was pre-specified from theory before running any test.

Restricting the primary regression to midday bars produces $\hat{\beta}_3 = 0.000812$, $z = 1.449$, $p = 0.147$, $N = 20{,}279$ — still not significant, but with a coefficient 2.2× larger than the full-sample estimate.

A more direct proxy for detector quality is the rolling standard deviation of signed order flow within the 30-bar lambda estimation window. Low variance indicates stable, representative order flow rather than a single large directional episode dominating the OLS estimate. Restricting the regression to the bottom tercile of this stability metric (threshold: 211 contracts signed-flow standard deviation) produces the gradient shown in Table 6.

Table 6. Stable Regime Conditions: $\hat{\beta}_3$ by Lambda Estimation Stability

Condition	N	$\hat{\beta}_3$	p-value
Full sample	55,634	0.000371	0.234
Stable lambda (bottom tercile)	18,355	0.001016	0.033
Unstable lambda (top tercile)	18,355	0.000331	0.664

The stable-condition $\hat{\beta}_3 = 0.001016$ is 2.7× the full-sample estimate and is statistically significant at $\alpha = 0.05$ ($z = 2.135$, $p = 0.033$). The gradient from stable to unstable is monotonic and strong. High-regime bars constitute 16.5% of the stable sample versus 12.1% of the full sample, consistent with stable estimation windows correctly classifying more bars as genuinely high-regime.

Two caveats apply. First, the stability threshold was derived from the in-sample distribution and is therefore post-hoc; the result is a finding but not a pre-registered test. Second, the stable conditions gradient does not generalize to the OOS period. The finding is therefore in-sample only, and its generalizability to other market environments or time periods is unknown.

The stable regime conditions result is presented as a genuine in-sample finding that motivates a specific future research direction. Whether removing the remaining circularity would strengthen, weaken, or eliminate the finding cannot be determined from the current data. Figures A2 and A3 in Appendix A show the average intraday profiles of Kyle's lambda and RegimeScore respectively. Lambda (Figure A2) is elevated and relatively stable at the open, reaches its session minimum around 12:00–12:30 ET, then rises and remains elevated through the afternoon before dropping sharply at the close. RegimeScore (Figure A3) is mechanically zeroed during the first hour of trading due to the 30-bar lambda warmup window, jumps to approximately 0.23 at 10:30 ET, and remains broadly stable through the session before dropping to zero at the 15:50 ET exclusion boundary. The flat midday RegimeScore profile is consistent with the rolling z-score standardization adapting to the lower-lambda midday environment rather than suppressing the score when lambda is at its session trough.

5.6. Subsample Stability

The primary regression is estimated separately on two activity sub-periods identified in Section 3.

Table 7. Subsample Stability: $\hat{\beta}_3$ by Activity Sub-Period

Subsample	N	$\hat{\beta}_3$	p-value
Full sample (May–Dec 2025)	55,634	0.000371	0.234
May–Sep 2025	35,001	0.000283	0.495
Oct–Dec 2025	20,633	0.000543	0.207

The null result is stable across both sub-periods. The directional pattern — higher $\hat{\beta}_3$ in the more volatile Oct–Dec period — is consistent with the regime detector having more genuine informed trading episodes to identify in elevated-volatility environments, but the difference between two non-significant p-values carries no inferential weight. The primary efficiency finding is a structural property of the sample, not a sub-period artifact.

5.7. Out-of-Sample Validation

The in-sample specification is applied without modification to the held-out 2026 period (2026-01-02 through 2026-03-06, N = 14,774, 46 trading days). No parameters are refit.

The full-period OOS result ($\hat{\beta}_3 = 0.000371$, $z = 2.768$, $p = 0.006$) is significant at $\alpha = 0.01$ and in the opposite direction of what the in-sample null would predict. Six candidate explanations were evaluated: HAC misspecification, regime distribution shift, volatility-driven lambda inflation, TFI distribution shift, random noise, and episode concentration. Residual autocorrelation at lags 6–10 ranges from +0.001 to +0.020, ruling out HAC misspecification. The OOS RegimeScore distribution is essentially identical to in-sample (mean 0.229 vs. 0.215, ratio 1.06×), ruling out structural regime elevation. Realized volatility in the OOS period is lower than in-sample (mean $|R|$ ratio 0.42×), ruling out volatility-driven lambda inflation. The OOS TFI distribution is nearly identical to in-sample (1.3% vs. 1.0% extreme bars), ruling out extreme order flow as a mechanical driver. A permutation test (1,000 random permutations of OOS forward returns) places the actual OOS $\hat{\beta}_3$ in the top 0.5% of the null distribution (permutation $p = 0.005$), ruling out random noise.

Monthly breakdown identifies the source: January and February 2026 are individually null. A rolling 10-day window shows $\hat{\beta}_3$ near zero through January and early February, then rising sharply to produce seven consecutive significant windows from February 26 through March 6, 2026.

Table 8. OOS Split: Early vs. Late Period

Period	$\hat{\beta}_3$	p-value	Return std (bps)	R^2
Early OOS (Jan–Feb 22; N=11,484)	0.000066	0.559	3.041	0.002
Late OOS (Feb 23–Mar 6; N=3,290)	0.000774	0.022	5.792	0.111

The OOS result is an episodic finding. The aggregate significance is driven by a specific two-week episode of elevated realized volatility in late February through early March 2026. January and February individually are null. The permutation test confirms the episode is not random noise, but no orthogonal conditioning variable explains or predicts it. The in-sample efficiency finding is not overturned.

6. Robustness

6.1. Lagged Regime Conditioning

As the most stringent robustness test, the primary regression replaces the contemporaneous RegimeScore_t and interaction $\text{TFI}_t \times \text{RegimeScore}_t$ with their one-bar lagged counterparts RegimeScore_{t-1} and $\text{TFI}_t \times \text{RegimeScore}_{t-1}$. This specification is fully predetermined — there is no simultaneity between any variable — and eliminates any residual circularity from the contemporaneous regime score sharing information with bar t's order flow.

$$\hat{\beta}_3 = -0.000152, \quad z = -0.476, \quad p = 0.634, \quad N = 55{,}634$$

The null result is unambiguous under this specification, implying that prior-bar regime information has no carry-forward predictive value for next-bar returns and that regime-conditioned adverse selection resolves entirely within one bar.

6.2. Regime Transition Dynamics

An additional specification tests whether TFI is differentially predictive at the first bar of a regime transition — the moment when RegimeScore crosses 0.5 from below — relative to sustained high-regime periods:

$$R_{t+1} = \alpha + \beta_1 \mathrm{TFI}_t + \beta_2 \mathrm{RegimeScore}_t + \beta_3(\mathrm{TFI}_t \times \mathrm{RegimeScore}_t) + \beta_4(\mathrm{TFI}_t \times \mathbf{1}[\mathrm{TransHigh}]_t) + \beta_5 R_t + \beta_6 \mathrm{TFI}_{t-1} + \varepsilon_t \quad (8)$$

The sustained high-regime coefficient is $\hat{\beta}_3 = 0.000143$ ($p = 0.684$) and the transition coefficient is $\hat{\beta}_4 = 0.000430$ ($p = 0.061$), across 2,354 transition bars (4.2% of sample). Two diagnostics confirm the apparent effect is mechanically inflated.

First, transition bars are split by the median RegimeScore delta at crossing (median $= 0.270$):

Table 9. Transition Bars Split by RegimeScore Delta at Crossing

Delta group	N transitions	$\hat{\beta}_4$	p-value
Small delta ($\leq$ median)	1,177	0.000046	0.875
Large delta ($>$ median)	1,177	0.000618	0.035

The large-delta coefficient is 13.4× the small-delta coefficient, extreme concentration inconsistent with genuine informed trader timing and strongly consistent with mechanical inflation.

Second, the transition threshold is varied:

Table 10. Transition Dynamics: Threshold Sensitivity

Threshold	Transition bars	$\hat{\beta}_4$	p-value
0.4	2,868	0.000200	0.263
0.5	2,354	0.000430	0.061
0.6	1,684	0.000362	0.190

No threshold produces a significant result. Both diagnostics confirm the transition specification is mechanically inflated and the null result is the correct interpretation.

6.3. Regime Detector Threshold Sensitivity

The detector validation regression (Section 4.4) classifies bars as high-regime when RegimeScore $>$ 0.5, pre-specified as the natural midpoint of the $[0, 1]$ logistic output. Table 11 reports $\hat{\beta}_3$ and the within-bar TFI-return amplification ratio at thresholds of 0.4, 0.5, and 0.6.

Table 11. Threshold Sensitivity: Detector Validation Regression ($\hat{\beta}_3$, p-value, amplification ratio)

Threshold	High-regime bars	$\hat{\beta}_3$	p-value	Amplification
0.4	17.9%	0.001226	< 0.001	2.061×
0.5	12.1%	0.001525	< 0.001	2.278×
0.6	6.6%	0.001937	< 0.001	2.553×

The validation finding is robust to threshold choice; $\hat{\beta}_3$ is statistically significant at $p < 0.001$ at all three values. The monotonic increase in $\hat{\beta}_3$ as the threshold tightens reflects the expected behavior of a continuous score: stricter thresholds select a smaller, more price-impact-concentrated subset of bars. This pattern does not indicate sensitivity to the specific threshold used.

6.4. Additive Combination Robustness

The multiplicative formulation of RegimeScore was adopted on theoretical grounds before formal regressions were run (Section 4.5). For transparency, the primary regression is re-estimated using an additive formulation: $\text{RegimeScore}_t = [\text{logistic}(z_{\lambda,t}) + \text{logistic}(z_{\text{TAR},t})]/2$. The two formulations classify substantially different fractions of bars as high-regime (RegimeScore > 0.5) for a structural reason: averaging two logistic outputs centered at 0.5 produces a distribution concentrated near 0.5, whereas the multiplicative product is far more selective.

Table 12. Additive vs. Multiplicative Combination: Primary Regression ($N = 55{,}634$)

Formulation	High-regime	$\hat{\beta}_3$	z-stat	p-value
Multiplicative (main)	12.1%	0.000371	1.191	0.234
Additive	43.1%	0.000187	0.451	0.652

The null result holds under both formulations. The primary efficiency finding is not sensitive to the choice of combination rule.

6.5. Lambda and TAR Window Sensitivity

The regime detector uses a 30-bar rolling OLS window for Kyle's lambda and a 5-bar rolling mean for the trade arrival rate, both pre-specified in Section 4.5. Tables 13 and 14 report primary regression $\hat{\beta}_3$ and p-value when each window is varied while the other is held fixed. The stable-conditions analysis from Section 5.5 is re-run at each lambda window length with the rolling signed-flow standard deviation window matched to the lambda estimation window.

Table 13. Lambda Window Sensitivity (TAR window fixed at 5 bars)

λ window	N (full)	$\hat{\beta}_3$	p (full)	N (stable)	$\hat{\beta}_3$	p (stable)
15	60,344	0.000826	0.006**	19,912	0.000597	0.215
30 (main)	55,634	0.000371	0.234	18,355	0.001016	0.033*
60	46,214	0.000599	0.080	15,241	0.001291	0.009**

Table 14. TAR Window Sensitivity (lambda window fixed at 30 bars, $N = 55{,}634$)

TAR window	$\hat{\beta}_3$	p-value
3	0.000389	0.243
5 (main)	0.000371	0.234
10	0.000414	0.157

The TAR window result is insensitive to window choice; $\hat{\beta}_3$ ranges from 0.000371 to 0.000414 and all three specifications are null ($p > 0.15$). The primary null result does not depend on the TAR smoothing parameter.

The full-sample result is sensitive to the lambda estimation window. At $\lambda = 30$, the primary null holds ($p = 0.234$). At $\lambda = 15$, the full-sample result is significant ($p = 0.006$); at $\lambda = 60$, it is marginal ($p = 0.080$). Observation counts differ because the rolling OLS warmup matches the window length, yielding 60,344 bars at $\lambda = 15$ and 46,214 bars at $\lambda = 60$. Shorter lambda windows increase the overlap between the lambda estimation sample and the current bar's signed order flow. Since signed order flow is the same input as TFI, this amplifies the first circularity layer documented in Section 4.5. The stable-conditions pattern is consistent with this explanation: at $\lambda = 15$ the full-sample significant result reverses to null under the stability filter ($p = 0.215$), whereas at $\lambda = 30$ and $\lambda = 60$ the opposite holds, with a null full-sample result and significant stable-conditions results ($p = 0.033$ and $p = 0.009$ respectively). The $\lambda = 15$ full-sample significance is therefore attributable to heightened circularity rather than independent evidence of regime-conditioned forward predictability. The stable-conditions finding from Section 5.5 is reinforced at $\lambda = 60$ ($\hat{\beta}_3 = 0.001291$, $p = 0.009$), carrying the same post-hoc caveat and the same out-of-sample null.

6.6. Expanded Announcement Exclusion Set

The primary specification excludes a 30-minute post-announcement window following each FOMC decision, CPI release, and NFP release (Section 4.3). Andersen et al. (2007) document that these three release types, together with PPI, advance GDP, and retail sales, generate the largest and most persistent intraday price responses in U.S. equity markets. To verify that the original exclusion set is not materially incomplete, the primary regression is re-run with all six release types excluded. For pre-market releases at 8:30 AM ET, the post-announcement window is shifted from the pre-RTH interval (8:30–9:00 AM, which contains no RTH bars) to the first 30 minutes of RTH (9:30–10:00 AM), capturing the opening period during which the pre-open release is priced in. Release dates reflect government-issued schedules, incorporating autumn 2025 shutdown-related delays where applicable.

The expanded exclusion set yields $\hat{\beta}_3 = 0.000326$ ($p = 0.291$, $N = 56{,}214$). The null result is unchanged. The observation count is 580 higher than the original sample rather than lower because the exclusion mechanism sets RegimeScore to zero in excluded windows rather than dropping rows from the regression; the new announcement-day opening windows convert a small number of `NaN` regime-score values to zero, which resolves downstream `NaN` in the lagged regime-score term and admits those bars to the regression. This is a minor sample composition artifact that does not affect the inferential conclusion. The primary null finding does not depend on the choice of announcement exclusion set.

7. Market Maker Implications

The formal results carry two directly actionable insights for market makers operating in ES futures.

The primary finding — no regime-conditioned forward predictability at any horizon from $T{+}1$ through $T{+}15$ — establishes that ES futures incorporate regime-conditioned order flow information within one 1-minute bar. A market maker who misses the within-bar adverse selection signal has no opportunity to adjust quotes before the price has fully moved. This underscores the importance of real-time rather than bar-level regime detection; the signal distinguishing informed from uninformed order flow is present and acting on prices within the bar, not across bars.

The contemporaneous characterization (Section 5.2) produces $\hat{\beta}_3 = 0.000617$ ($p = 0.078$), which is not statistically significant, and retains irreducible TFI-return confounding within bar t. No empirically grounded quantitative calibration of adverse selection amplification can be derived from the current data. The dominant explanation for the marginal p-value is residual regime autocorrelation (lag-1 autocorrelation = 0.806), not genuine causal amplification.

The strongest in-sample result is the stable-conditions gradient documented in Section 5.5. When the lambda estimation window is stable, defined as the bottom tercile of rolling signed-flow standard deviation, the regime-conditioned TFI interaction coefficient is $\hat{\beta}_3 = 0.001016$ ($p = 0.033$), statistically significant and 2.7× the full-sample estimate. Three caveats preclude any directional application of this result. First, the stability threshold is derived post-hoc from the in-sample distribution; no theoretical prior determines the bottom tercile as the operationally relevant cutoff. Second, the finding does not replicate in the held-out out-of-sample period. Third, the stability metric itself remains partially circular with the regime detector: rolling signed-flow standard deviation shares the same aggressor-side input as TFI, and low-stability bars are mechanically associated with low TFI variance in a way that may inflate the interaction term in the stable subset independently of any genuine signal. Whether replacing the signed-flow proxy with an orthogonal measure, such as bid-ask spread stability, depth constancy, or cancellation rate, would strengthen, weaken, or eliminate the finding cannot be determined from the current data. The stable-conditions gradient motivates a specific future investigation, but the three caveats documented above preclude any current directional application.

Net-return magnitude. A back-of-envelope calculation characterizes the noise-level magnitude of the regime-conditioned signal in the high-regime/high-|TFI| cell. In high-regime bars (RegimeScore > 0.5, $N = 6{,}719$ in-sample), mean $|\text{TFI}| \approx 0.08$ (computed from the in-sample distribution). The full-sample coefficient $\hat{\beta}_3 = 0.000371$ is a null result ($p = 0.234$); the corresponding calculation at mean $|\text{TFI}| = 0.08$ and representative RegimeScore $= 0.75$ yields:

$$0.000371 \times 0.08 \times 0.75 \;\approx\; 0.000022 \;\approx\; 0.22 \text{ bps}$$

This figure is not a signal magnitude estimate. It is an upper bound on the noise-level magnitude consistent with the null result, establishing that even the maximum plausible regime-conditioned effect is economically negligible. Using the stable-condition coefficient $\hat{\beta}_3 = 0.001016$, the same calculation yields $\approx$0.61 bps; this is an in-sample estimate subject to the three caveats stated above and should not be interpreted as a plausible signal estimate outside the in-sample context. Round-trip transaction costs are approximately 0.774 bps. The null upper bound (0.22 bps) falls below transaction costs, confirming the primary result is economically negligible. The stable-condition in-sample estimate (0.61 bps) is approximately 79% of round-trip costs before accounting for adverse selection risk at the point of execution, where the market maker is transacting against informed flow. High-regime bars represent 12.1% of the regression sample (6,719 of 55,634 bars), yielding approximately 40 high-regime bars per trading day; this frequency is sufficient to amortize fixed costs but does not alter the gross-vs-net shortfall. No quantitative calibration of adverse selection amplification can be derived from these figures for the reasons stated above.

8. Future Research

Two research directions are motivated directly by the current findings and their limitations.
Direction 1: Orthogonal regime detector with limit order book data. The two layers of circularity documented in this paper — at the detector construction level and the regression interaction level — are both irreducible with trades-only single-instrument data. A regime detector derived from limit order book dynamics (bid-ask spread changes, depth imbalance, cancellation-to-trade ratio) would share no informational content with TFI and would permit a genuinely non-confounded test of the regime-conditioned forward predictability hypothesis. The stable regime conditions result ($\hat{\beta}_3 = 0.001016$, $p = 0.033$ under stable estimation conditions) motivates this direction but does not predict its outcome. An LOB-based detector with an orthogonal stability criterion could determine whether the stable conditions effect strengthens, weakens, or disappears when circularity is removed; a null result under an orthogonal detector would itself be an important finding, confirming that the in-sample result was circularity-driven. Two sub-questions follow only if the effect survives: does it persist at sub-minute resolution, and does it survive two-sided transaction costs?
Direction 2: p% intra-bar amplification framework. If a future study finds significant contemporaneous regime amplification with an orthogonal detector, the current framework can be extended to characterize the within-bar dynamics of adverse selection. Specifically, TFI computed at fixed elapsed-time fractions within each bar (10%, 20%, ..., 90% bar completion) could be regressed on bar-end returns conditioned on lagged RegimeScore_{t-1}, yielding an empirical weight schedule θ_p describing how information content accumulates within the bar as a function of elapsed time and regime state. This framework provides the bridge from a statistically significant regime-TFI relationship to a practically implementable real-time adverse selection signal. It is constructible from current tick data but deferred pending a significant and non-confounded contemporaneous finding.

A. Supporting Figures

Figures A1–A5 provide visual support for claims made in the main text. No inferential claim in Sections 4–7 depends on visual inspection of these figures.

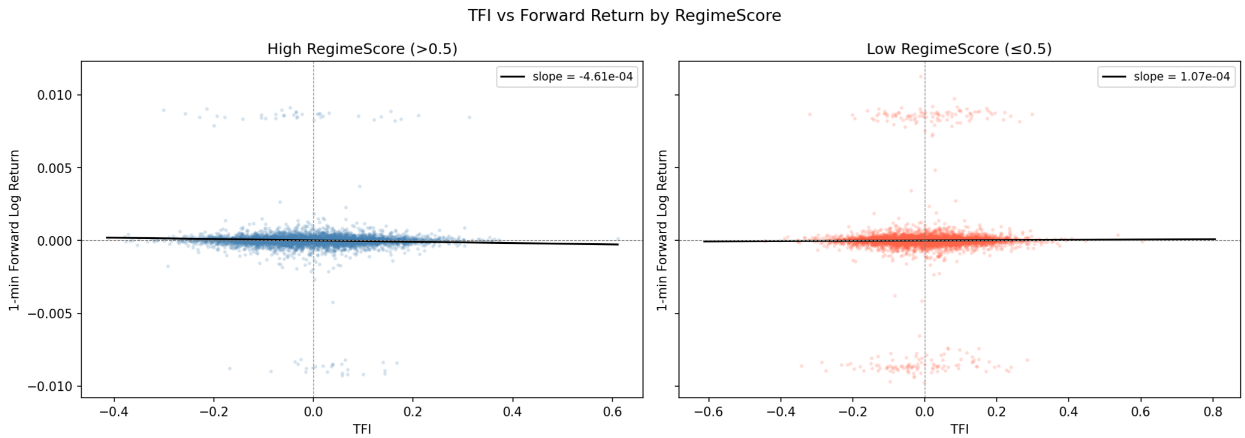

Figure A1. TFI versus one-bar-ahead log return, stratified by regime state. Two side-by-side scatter panels (up to 5,000 sampled points each). Left: high-regime bars (RegimeScore > 0.5). Right: low-regime bars (RegimeScore ≤ 0.5). The OLS slope is -4.61×10^{-4} in the high-regime panel and $+1.07 \times 10^{-4}$ in the low-regime panel; neither is economically distinguishable from zero. The sign reversal in the high-regime panel is consistent with the mechanical co-elevation argument of Section 4.5. Visual analog of Table 2.

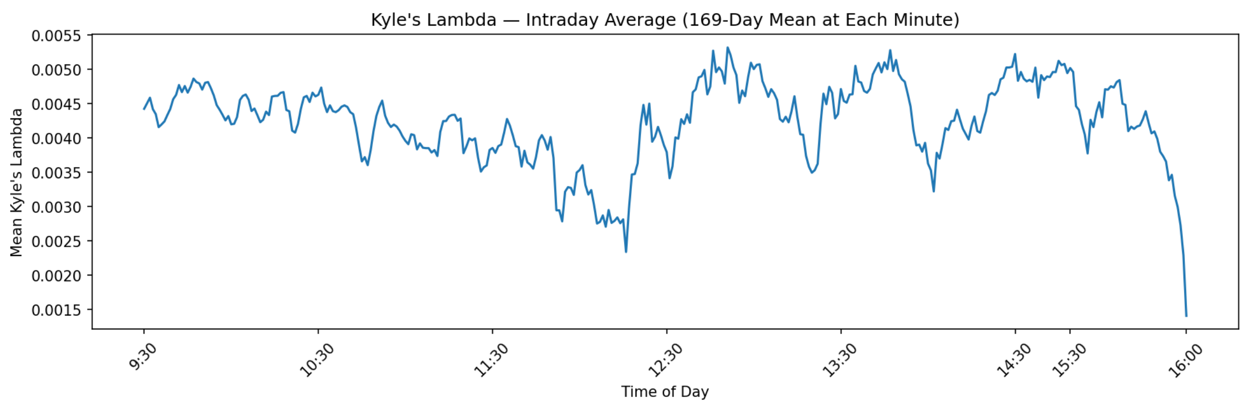

Figure A2. Average intraday Kyle's lambda by minute, averaged across 169 in-sample trading days. Lambda is elevated and relatively stable near the open (≈0.0045), declines to its session minimum around 12:00–12:30 ET (≈0.0024), then rises and remains elevated through the afternoon (≈0.0048–0.0053) before dropping sharply at the close as MOC-dominated flow suppresses price impact. The midday trough is the period of most stable lambda estimation, directly supporting the analysis in Section 5.5.

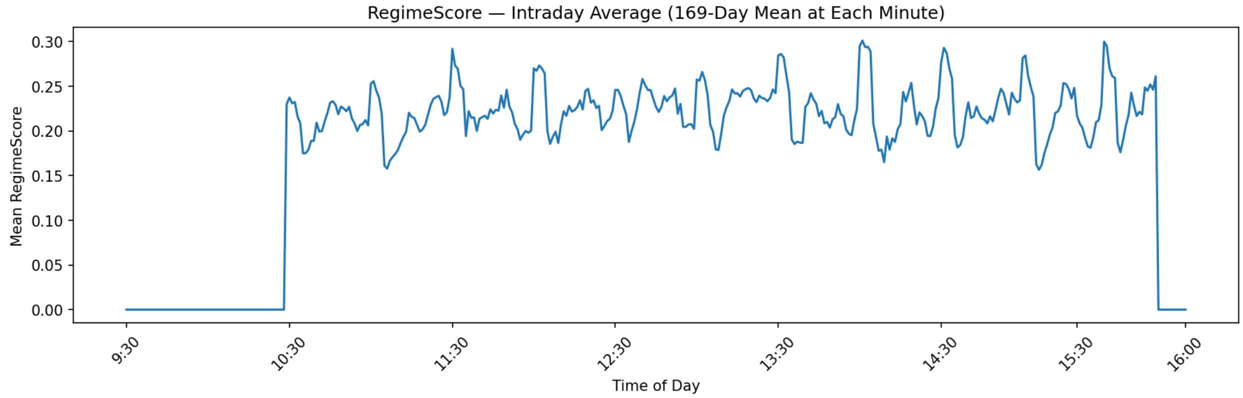

Figure A3. Average intraday RegimeScore **by minute,** averaged across 169 in-sample trading days. RegimeScore is mechanically zero from 09:30 through approximately 10:30 ET — the 30-bar lambda warmup window — before jumping sharply to ≈0.23 and remaining broadly stable through the session. The hard drop to zero at ≈15:50 ET reflects the end-of-session exclusion window. Unlike the lambda intraday profile (Figure A2), RegimeScore does not exhibit a midday trough; the rolling z-score standardization causes the score to reflect relative rather than absolute lambda levels.

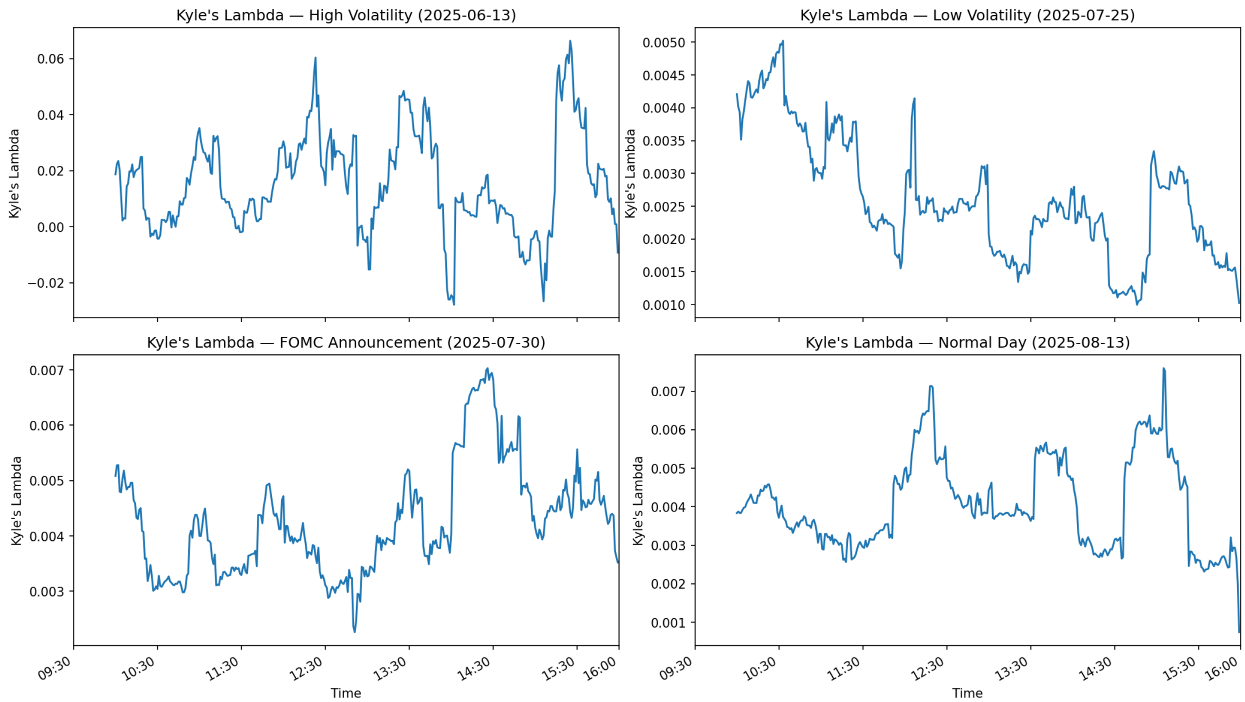

Figure A4. Intraday Kyle's lambda for four representative trading days. *Top left:* High-volatility session (2025-06-13), showing large-amplitude swings and negative lambda readings reflecting extreme directional price moves. *Top right:* Low-volatility session (2025-07-25), showing a compressed, stable profile with lambda ranging narrowly between ≈0.001 and 0.005. *Bottom left:* FOMC announcement day (2025-07-30), showing low and stable lambda through the morning, followed by a sharp rise around 14:00–14:30 ET coinciding with the announcement window and sustained elevation through the close. *Bottom right:* Normal session (2025-08-13), showing a profile similar in scale to the low-volatility day with moderate intraday variation. Together these four panels illustrate how lambda responds to qualitatively different market conditions and motivate the FOMC exclusion window design.

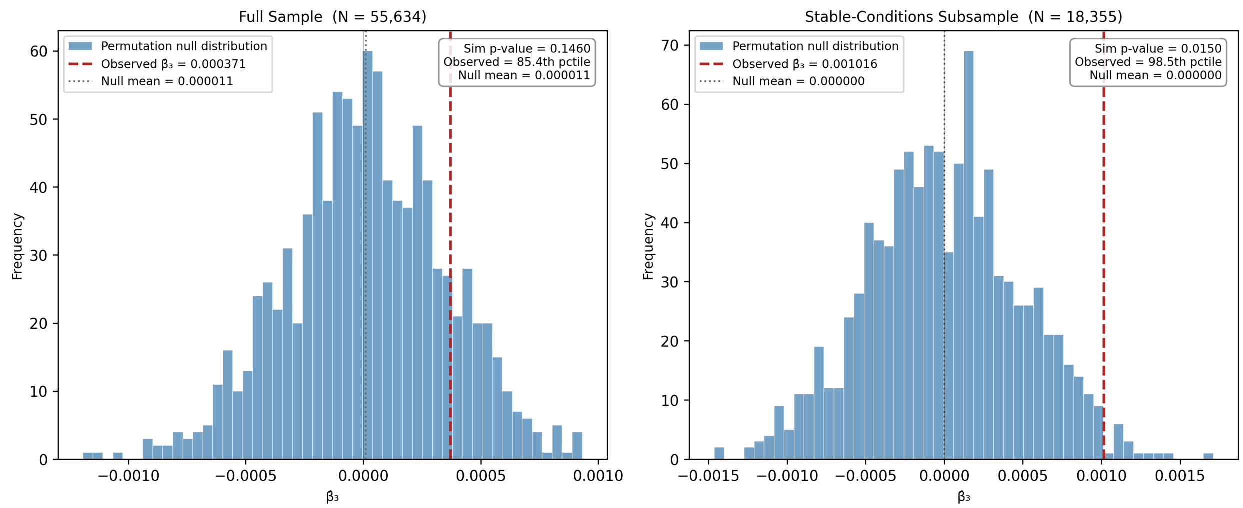

Figure A5. Permutation null distribution of $\hat{\beta}_3$ (1,000 permutations, seed = 42). Each panel shows the histogram of $\hat{\beta}_3$ recovered from 1,000 permutations of forward returns R_{t+1}, with TFI_t and RegimeScore_t held fixed. The red dashed line marks the observed $\hat{\beta}_3$; the gray dotted line marks the null mean. *Left:* Full sample ($N = 55{,}634$). The null mean is +0.000011, confirming upward bias from mechanical co-elevation. The observed $\hat{\beta}_3 = 0.000371$ falls at the 85.4th percentile (simulation p-value = 0.146), consistent with a null result. *Right:* Stable-conditions subsample ($N = 18{,}355$). The null mean is 0.000000 and the observed $\hat{\beta}_3 = 0.001016$ falls at the 98.5th percentile (simulation p-value = 0.015), confirming the stable-conditions result is not a mechanical artifact of upward bias. See Section 4.5.

References

Ahern, K. R. (2018). Do proxies for informed trading measure informed trading? Evidence from illegal insider trades. *Review of Asset Pricing Studies*, 10(3):397–440.

Andersen, T. G., Bollerslev, T., Diebold, F. X., and Vega, C. (2007). Real-time price discovery in global stock, bond and foreign exchange markets. *Journal of International Economics*, 73(2):251–277.

Cont, R., Kukanov, A., and Stoikov, S. (2014). The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88.

Dufour, A. and Engle, R. F. (2000). Time and the price impact of a trade. *Journal of Finance*, 55(6):2467–2498.

Easley, D., López de Prado, M. M., and O'Hara, M. (2012). Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies*, 25(5):1457–1493.

Glosten, L. R. and Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1):71–100.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335.

O'Hara, M. (1995). *Market Microstructure Theory*. Blackwell, Oxford.

Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance*, 39(4):1127–1139.